

Comlinear Corporation  
SPICE Model Diskette (MMD.05)  
4800 Wheaton Drive  
Fort Collins, CO 80525

303-226-0500  
1-800-776-0500

Comlinear Corporation is a manufacturer and supplier of high-performance analog signal processing components. Comlinear's broad signal conditioning product line includes high-speed hybrid and monolithic operational amplifiers, buffers, video amplifiers, multiplexers, automatic gain control integrated circuits, track/hold amplifiers, and analog-to-digital converters. Comlinear continues as a leader in developing products offering exceptional performance, speed, quality, reliability and service.

## INTRODUCTION

This diskette is a collection of Spice compatible models for Comlinear amplifiers. For additional information about SPICE Models supporting existing or new products, Comlinear customers can contact Comlinear Corporation directly at 1-800-776-0500. The SPICE Models found on this disk are created for use on an IBM compatible computer using analysis programs that accept Spice formats. Comlinear assumes no responsibility for designs created from these SPICE Models. These SPICE Model files model typical performance at room temperature. Before designs are released to production, Comlinear suggests that topologies be verified by prototyping the circuit. The part-to-part and over-temperature performance variations of Comlinear amplifiers are specified in current data sheets (see Reference 1). For more detailed information concerning the use of these SPICE Models, please refer to Comlinear's Application Note OA-18.

The changes from the last SPICE Model diskette version are listed in this table:

TABLE I. UPDATES TO SPICE MODEL DISKETTE

| =====      |                                                                            |
|------------|----------------------------------------------------------------------------|
| CLC109.CIR | A new SPICE Model.                                                         |
| CLC111.CIR | A new SPICE Model.                                                         |
| CLC402.CIR | An updated version that gives better noise, dc bias, and CMRR performance. |
| CLC426.CIR | A new SPICE Model.                                                         |
| CLC428.CIR | A new SPICE Model.                                                         |
| CLC431.CIR | A new SPICE Model.                                                         |
| CLC432.CIR | A new SPICE Model.                                                         |
| CLC502.CIR | An updated version that gives better noise, dc bias, and CMRR performance. |

SPICE Model files found on this disk are:

TABLE II. SPICE MODEL SUBCIRCUIT FILES ON THIS DISKETTE

| File Name  | Description                                                |
|------------|------------------------------------------------------------|
| =====      |                                                            |
| CLC109.CIR | A Low-Power, Wideband, Closed-Loop Buffer.                 |
| CLC111.CIR | A Very Wideband, Ultra-High Slew Rate, Closed-Loop Buffer. |

CLC400.CIR A Wideband, Low-Gain Monolithic Current Feedback Op Amp with Fast Settling, (.05% in 12 ns), Low Power and an Input Offset Adjustment Pin.

CLC401.CIR A Wideband, High-Gain Monolithic Current Feedback Op Amp with Fast Settling (.01% in 10 ns) and Low Power.

CLC402.CIR A Low-Gain Monolithic Current Feedback Op Amp with Fast 14-bit Settling (.0025% in 25 ns) and Low Power.

CLC404.CIR A Wideband Monolithic Current Feedback Op Amp with High Slew Rate.

CLC406.CIR A Wideband Low-Cost, Low-Power Monolithic Current Feedback Op Amp.

CLC409.CIR A Very Wideband, Low Distortion Monolithic Current Feedback Op Amp.

CLC410.CIR A Video Monolithic Current Feedback Op Amp with disable, Fast Settling (0.05% in 12 ns) and an Input Offset Adjust Pin.

CLC414.CIR A Quad, Low-Power Monolithic Current-Feedback Op Amp.

CLC415.CIR A Quad Wideband Monolithic Current Feedback Op Amp.

CLC420.CIR A High-Speed, Unity Gain Stable Monolithic Voltage Feedback Op Amp.

CLC425.CIR An Ultra Low-Noise, Wideband Monolithic Voltage Feedback Op Amp with Current Supply Adjust.

CLC426.CIR An Ultra Low-Noise, Wideband Monolithic Voltage Feedback Op Amp with Current Supply Adjust and External Compensation.

CLC428.CIR An Ultra Low-Noise, Wideband, Dual Monolithic Voltage Feedback Op Amp

CLC430.CIR A Wideband Monolithic Current Feedback Op Amp with disable and +/-5V to +/-15V supply capability.

CLC431.CIR A Dual, Wideband Monolithic Current Feedback Op Amp with high slew rate.

CLC432.CIR A Dual, Wideband Monolithic Current Feedback Op Amp with disable and +/-5V to +/-15V supply capability.

CLC501.CIR A High-Speed Output Clamping Monolithic Current Feedback Op Amp for high gains.

CLC502.CIR A High-Speed Output Clamping Monolithic Current Feedback Op Amp with Fast 14-bit Settling (0.0025% in 25 ns) for low gain.

CLC505.CIR A High-Speed, Programmable-Supply Current, Monolithic Current Feedback Op Amp.

CLC520.CIR A Monolithic Amplifier with Voltage Controlled Gain (AGC).

CLC522.CIR A Monolithic Wideband Variable Gain Amplifier.

CLC532.CIR A High-Speed, 2:1 Analog Multiplexer with fast 12-bit settling (0.01% in 17 ns), low noise, low distortion and adjustable noise bandwidth.

SPICE Models are found in the root directory of the data disk supplied.

START UP INSTRUCTIONS

Make a backup copy of the SPICE files contained on this disk to another floppy. Copy all SPICE Model files of interest to a library on the hard disk. If the library directory is not in the SPICE program's path, the user should set that path in the autoexec.bat for easier excess. The .INC statement in PSpice should be used in the simulation file to include the SPICE Models subcircuit.

Example       " .INC CLC400.CIR"

## AMPLIFIER SPICE MODELS

These SPICE Model files are written in ASCII file format for IBM-compatible PC's. They are compatible with Spice 2G simulators.

Comlinear amplifier SPICE Models

are written in a subcircuit format for easy incorporation into larger circuits. A listing of any amplifier subcircuit may be obtained by printing its CLC\*.CIR file to a local printer.

The subcircuit node assignments match the device pin-outs as shown in the individual device data sheets. An example is an 8 pin op amp.

|                |                         |
|----------------|-------------------------|
| * Connections: | NON-INVERTING INPUT PIN |
| *              | INVERTING INPUT PIN     |
| *              | OUTPUT                  |
| *              | +VCC                    |
| *              | -VCC                    |
| *              |                         |
| .SUBCKT (NAME) | 3 2 6 7 4               |

Some schematic capture software packages require a different pin connection order than what Comlinear uses. Changing the pin order in the .SUBCKT statement will not affect the SPICE Model performance.

## PERFORMANCE RESULTS

When substitutions of current feedback op amps are made for voltage feedback op amps, results may not be acceptable. Refer to Comlinear's application note OA-13 for a tutorial on current feedback op amp design.

## PARAMETERS MODELED

The following typical performance parameters are modeled by the SPICE Models.

### DC Effects

- \* VIO, IBI, IBN
- \* Supply current vs. Supply voltages
- \* Common Mode Input/Output Voltage range
- \* Load current from supplies
- \* CMRR

### AC Effects

- \* Frequency response vs. gain & load.
- \* Open loop gain & phase
- \* Noise
- \* Small signal Input/Output Impedance

#### Time Domain

- \* Rise / fall times
- \* Slew Rates

#### Special Features (where applicable)

- \* Output Clamping
- \* Supply Current Adjustment
- \* Offset voltage adjust
- \* Disable / Enable times
- \* External Compensation

#### PARAMETERS NOT MODELED

- \* Differential gain and phase
- \* PSRR
- \* Harmonic distortion
- \* Fine scale settling performance
- \* Thermal tail
- \* Overdrive recovery time (Except for the CLC501 and the CLC502)
- \* Variation in performance vs. temperature
- \* Part-to-part performance variation

#### NOTICE

The information provided within this file is believed to be reliable. Comlinear assumes no responsibility for alterations, omissions or inaccuracies. Comlinear also assumes no responsibility for the use of this information, and all use of such information shall be entirely at the user's risk. Comlinear does not grant licenses or patent rights to any of the circuits described within this document.

#### REFERENCES

- 1) Comlinear Databook of standard products.